



Feature Article

Anti-fouling properties of microstructured surfaces bio-inspired by rice leaves and butterfly wings



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ABSTRACT

Material scientists often look to biology for new engineering solutions to materials science problems. For example, unique surface characteristics of rice leaves and butterfly wings combine the shark skin (antifouling) and lotus leaf (self-cleaning) effects, producing the so-called rice and butterfly wing effect. In this paper, we study antifouling properties of four microstructured surfaces inspired by rice leaves and fabricated with photolithography and hot embossing techniques. Anti-biofouling effectiveness is determined with bioassays using *Escherichia coli* whilst inorganic fouling with simulated dirt particles. Antifouling data are presented to understand the role of surface geometrical features resistance to fouling. Conceptual modeling provides design guidance when developing novel antifouling surfaces for applications in the medical, marine, and industrial fields.

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1. Introduction

Engineers and scientists often seek inspiration from biology to solve engineering problems. Developing a deeper appreciation of living nature can lead to bioinspired solutions for many industries. For instance, examples include “antifouling and low drag” surfaces inspired by microstructured shark skin as well as “self-cleaning” transparent coatings inspired by the hierarchical superhydrophobic and low adhesion lotus leaf [1–6].

Two engineering problems facing engineers are biofouling and inorganic fouling of surfaces. Biofouling is the accumulation of undesired organisms on surfaces, with biofilms created by microorganisms and macroscale biofouling (macrofouling) created by organisms (for instance, barnacles). Inorganic fouling includes deposits of dirt, corrosion, crystals, suspended particles, oil, and ice. The type and extent of fouling depends on the environment, inorganic deposits, and organisms [7].

In biomedical contexts, fouling typically takes the form of biofilms [8–12], whereas marine and industrial biofouling typically includes macrofouling [13–16] and inorganic fouling [17–20]. The term fouling describes both biofouling and inorganic fouling, where antifouling is the prevention of any biological or inorganic fouling. Conversely, self-cleaning is the removal of fouling already deposited on a surface.

Fouling can create significant health risks and loss of resources in medical, marine, and industrial applications. Examples of the various types and problems with fouling are shown in Table 1, highlighting the many applications susceptible to biofouling and inorganic fouling. Fig. 1 shows medical biofouling can occur in regions such as prosthetic implants, biosensors, catheters, dental implants, and medical equipment. Biofouling may lead to implant rejection, malfunction of biosensors, and spread of infectious diseases [9–12]. Biofilms on medical devices including a ventilation tube, needless connector, and pacemaker wire are shown in Fig. 2. Biofouling is commonly associated with marine environments where noticeable marine growth appears on ships and

underwater structures. This magnifies ship drag, corrosion, fuel consumption, and engine stress [15,20–23]. Industrial fouling occurs in areas such as power plants, water treatment systems, and food or beverage industries, as shown in Fig. 3. Issues include pipe blockage, decreased reverse osmosis (RO) membrane flux, contaminated water, and reduced heat exchanger efficiency [12,14,17].

Inorganic fouling creates concern in many industries with decreased performance in applications ranging from power production to consumer products. For example, the interior surface of a heat exchanger tube from a nuclear reactor is shown in Fig. 4, and corrosion deposits are readily apparent, which leads to inefficient performance [24] (<http://www.hcheattransfer.com>). Inorganic fouling also includes undesirable fingerprints on transparent surfaces such as electronic touch screens (mobile phones, tablets, music players, GPS devices), eyeglasses, computer screens, and window glass [25]. Furthermore, inorganic fouling includes ice on airplane wings, helicopter blades, oil platforms, power lines, locks and dams, and wind turbines [26,27].

Biology offers multiple examples of physical–chemical fouling control approaches that might be mimicked in synthetic structures singly or in combination [2,6,7]. Physical self-cleaning controls include low drag, low adhesion, wettability, microtexture, grooming, and sloughing; and chemical controls include various secretions [7]. Fouling is both prevented and removed with the aforementioned physical and chemical controls. In the ambient environment plants and insects employ these controls; and in the marine environment plants, corals, and fish employ them. Specific antifouling examples range from low drag shark skin [28–37] to self-cleaning lotus leaf [38–41].

Nature’s self-cleaning methods are commonly associated with the lotus effect that combines superhydrophobicity and low adhesion, but can also include superhydrophilicity, as demonstrated with pitcher plant. Self-cleaning of lotus leaves occurs when water beads up and rolls off of leaf surfaces carrying contaminants [2,4,42,43]. Conversely, pitcher plant self-cleaning occurs when water uniformly spreads and slides off carrying contaminants [4,44,45]. Similar self-cleaning mechanisms are possible with a variety of liquids, depending on the surface’s ability to repel or attract the liquids.

Producing effective low cost antifouling surfaces may be possible by embossing patterns onto flexible adhesive backed polymer films. Such films could be applied to flat or curved substrates such as oil pipeline interiors and ship hulls. Hot embossing may be conducted with a master mold that is heated and pressed into the polymer film [46–48] with techniques depending on the type of master mold and molding materials (i.e. composition, thickness, feature sizes). Master molds are often fabricated from silicon [46,49] or metal [50] with the embossed sheet commonly polymethylmethacrylate (PMMA) [46,48,51–54]. Embossing has been shown to fabricate various structures yielding sub-micron scale features [52,55].

Several bioassay antifouling experiments have been performed with different microorganisms and microstructured surfaces – using pillars, ribs, and a combination thereof with various diameters, widths, lengths, heights, and pitches [31,56–61]. Furthermore, the Sharklet AF™ micro-sized topographies based on shark skin riblets have been the subject of many studies [31,58,59]. Laboratory research suggests that samples with appropriately sized

Table 1

Fields susceptible to biofouling including common examples (adapted from Bixler and Bhushan [7]).

Type	Problems
<i>Medical</i>	
Orthopedic implant	Removal due to infection
Respirator	Ventilator associate pneumonia
Contact lens	Eye infection
Catheter	Urinary tract infections
Hemodialysis	Infectious break-outs
Teeth/dental implant	Periodontal disease, gingivitis
Biosensor	Failure from fibrous encapsulation
<i>Marine</i>	
Ship hull	Increased fuel consumption
Ship engine	Increased stress from extra drag
Marine platform	Increased marine structure load/stress/fatigue
Metal	Increased biocorrosion
<i>Industrial</i>	
Membrane	Reduced flux
Fluid flow	Frictional loss in pipes
Drinking water	Pathogens in potable water
Food, paper, and paint	Food spoilage and worker health risks
Metal cutting fluid	Filter blockage and worker health risks

topographies prevent colonization by various microorganisms including green algae *Ulva* spores [31,59], porcine cardiovascular endothelial cells [31], and *Balanus Amphitrite* cyprids [58]. Such microorganisms have been exposed to samples and the antifouling effectiveness is determined by counting settled cells using light [31] and epifluorescence microscopy [59]. Up to 85% spore settlement reduction is reported with micro-sized ribs when compared to flat control samples [31]. This reduction is attributed to the micro-sized feature spacing that is slightly smaller than the microorganisms under investigation. The so-called engineered antifouling microtopographies are designed to target specific microorganisms based on their shape and size [58,59]. Other parameters that were investigated include the effect of roughness with the so-called engineered roughness index (ERI) [59] as well as wettability. Superhydrophobic and low adhesion surfaces are believed to provide antifouling through self-cleaning Lotus effect mechanisms [31,32,62].

To demonstrate the feasibility of these approaches for scale-up on synthetic surfaces, Bixler and Bhushan [48,63,64] reported that rice leaves and butterfly wings combine the desirable shark skin

(antifouling) and lotus leaf (self-cleaning) effects; creating the so-called rice and butterfly wing effect. Water droplets moving effortlessly in the direction indicated on rice leaf and butterfly wing surfaces are shown in Fig. 5. These unique surfaces exhibit anisotropic flow, water repellency, and low adhesion properties; which are believed to promote antifouling and self-cleaning. It is reported that sinusoidal grooves in rice leaves and aligned shingle-like scales in butterfly wings provide the anisotropic flow to direct water droplets. Hierarchical structures consisting of micropapillae superimposed by waxy nanobumps in rice leaves and microgrooves on top of shingle like scales in butterfly wings provide the superhydrophobicity and low adhesion [48,63,64].

In the current study, rice leaf inspired samples have been fabricated using photolithography and soft lithography methods in the laboratory and experimentally evaluated for their antifouling effectiveness. In order to demonstrate the feasibility of scaled-up manufacturing, samples from a hot embossing study were selected for antifouling experiments. Results are discussed and conceptual models are presented suggesting the antifouling effectiveness of surfaces inspired by the rice and butterfly wing effect. This paper

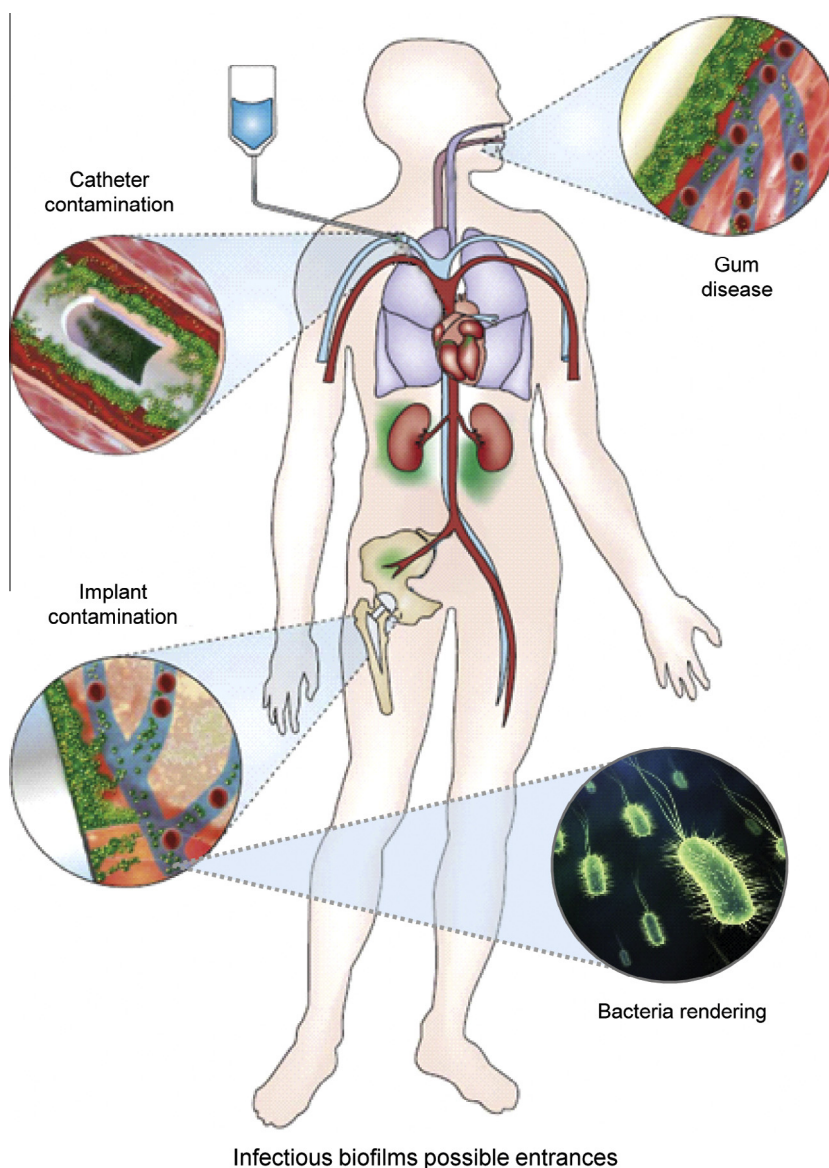


Fig. 1. Schematic highlighting areas and entry points of the body susceptible to infectious microorganisms leading to biofilms [7].

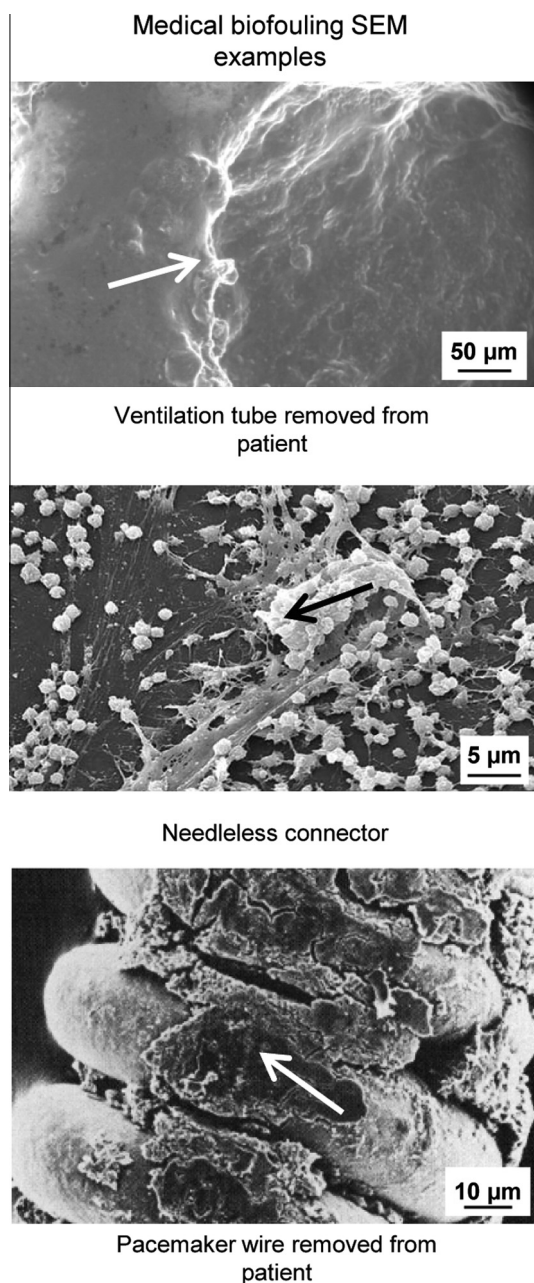


Fig. 2. Medical device biofilm examples [7]. Images include a ventilation tube (top) [70], needless connector (middle) (adapted from Janice Carr), and pacemaker wire (bottom) [67]. Such biofilms may lead to infections that require device removal since antibodies are often ineffective.

provides an overview of antifouling work as well as results from new anti-biofouling and previous anti-inorganic fouling research using rice leaf inspired samples.

2. Biofouling and inorganic fouling formation

In this section, we explain biofouling and inorganic fouling with regards to their type, formation, and morphology characteristics (see also review by Bixler and Bhushan [7]).

2.1. Biofouling formation

Biofouling begins when microorganisms become stationary on a particular substrate. In biofilms, microorganisms adhere to one

another and the substrate with an adhesive called the extracellular polymeric substance (EPS). The biofilm continues to grow and become more diverse by attracting additional microorganisms through chemical “messages”. The general principles of biofilm formation and factors leading to the settlement on hard surfaces are similar in medical, marine, and industrial applications [7,8,12,14,15,20,65,66].

Colonization is the process of organisms collecting and growing on a surface, and the morphology is characterized by the thickness, density, structure, composition, bioadhesive strength, and weight of fouling organisms [12,16,67–70]. In the marine environment, this usually starts with the formation of a biofilm that attracts larger macrofoulers [8,15,16,66]. For instance, tubeworms prefer settling on biofilms, but bryozoans and barnacles do not require a biofilm [34]. Fig. 6 illustrates the five-stage colonization process, which includes initial attachment, irreversible attachment, initial growth, final growth, and dispersion. Initial attachment starts the colonization process, which begins within days to a few weeks in a nutrient rich environment. The biofilm covered surface then attracts other organisms that may have been previously deterred. Initial attachment of microorganisms is reversible, but once they secrete the EPS, the bond becomes irreversible. This permanent attachment allows initial growth, final growth, and dispersion [8,15,65,66].

Underwater environments are ideal for biofouling since currents deliver nutrients and carry away wastes, promoting colonization by planktonic and sessile organisms. Biofouling growth rates depend on the organism, substrate, flow velocity, shear stress, and temperature [15,71,72]. Organism transportation to a surface is either passive from the current or active from the propagule, juvenile, or adult organism. Active mechanisms include electrostatic repulsion, Brownian motion, turbulent pulsations, and cell outgrowths [15,34].

As far as the tenacity of the biofouling is concerned, bioadhesion plays an important role [8,15,65,66]. Bioadhesion is the adhesion strength of biofouling on a hard surface. This depends on the organism type, substrate, and separating fluid [73], due to influences of electrostatic forces and surface wettability [42,74–76]. Biofilm bioadhesion is a two-stage process, starting with the initial attachment and then the irreversible attachment. Initial attachment is controlled by a physical adhesion between the microorganism and the substrate. Also known as adsorption, the initial colonists attach to a surface through weak, reversible van der Waals bonds, which are slightly stronger than electrostatic repulsive force. Irreversible attachment is accomplished with secretion of the EPS, which exhibits a sponge-like matrix. This adhesive permanently bonds the microorganisms to one another and collectively to the surface [8,15,77].

2.2. Inorganic fouling formation

Inorganic fouling is composed of non-biological particles that may form in addition to or independently from biofouling. Particles originate from corrosion, crystallization, suspended particles, oil, and ice. For instance, salts from aqueous solutions crystallize and deposit on surfaces. Other deposits may result from minerals found in water such as magnesium, calcium, and barium [7,12,14,17].

Types of inorganic fouling include particulate, freeze, and gas stream particulate. Particulate fouling occurs when suspended solid particles deposit onto a heat transfer surface [78]. Deposition of crystals from freeze fouling occurs in locations such as cold region oil pipelines when waxy hydrocarbons contact cold pipe walls. Gas stream particulate fouling occurs in gas lines, reactors, combustion chambers, and heat exchangers. This includes mineral, organic, and inorganic particles, which are common in oil or gas combustion systems [79]. Biofouling may initiate inorganic fouling, where

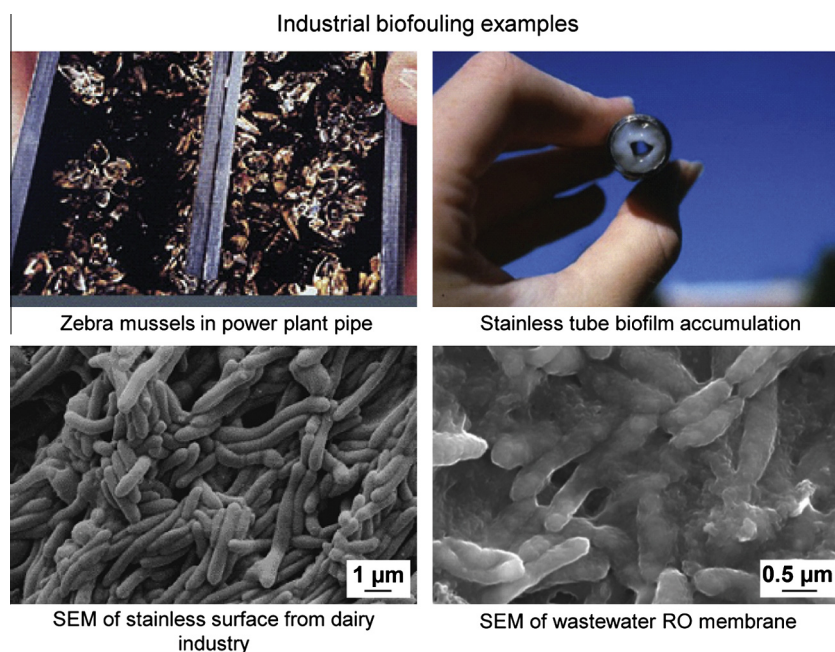


Fig. 3. Industrial biofouling examples [7]. Zebra mussels (*Dreissena polymorpha*) clogging power plant intake pipe (top left) (photo courtesy of Peter Yates), biofilm accumulation in a stainless steel tube (top right) [24], biofilm (*B. cereus*) on dairy industry stainless steel plate (bottom left) [94], and biofilm (*P. aeruginosa*) on reverse osmosis (RO) membrane (bottom right) [121].

biocorrosion causes the formation of corrosion particles. Such fouling is prevalent in boilers, cooling condensers, desalination plants, food processing equipment, geothermal plants, and oil production equipment [19]. Heat exchangers can develop hard deposits called “scale” or more porous deposits such as “sludge” [18].

3. Biomimetic antifouling strategies

In this section, we explain mechanisms behind shark skin and lotus leaf effects, which leads to the rice and butterfly wing effect.

3.1. Shark skin and lotus effects

In living nature, many plants and animals thrive due to their antifouling and self-cleaning characteristics [4,7,33,34,37–39,42,80,81]. For example, the skin of fast swimming sharks exhibits antifouling and self-cleaning properties. It is believed that sharks require low drag skin in order to swim quickly to catch prey, and that antifouling is necessary in order to maintain the low drag. Lotus leaves also exhibit antifouling with self-cleaning properties. It is believed that since lotus plants thrive in humid, marshy environments, self-cleaning prevents unwanted fouling, which may inhibit photosynthesis.

Shark skin microstructured riblets on dermal denticles reduce drag and prevent biofouling, as compared to other aquatic species. For instance, whales are often covered with barnacles; however sharks remain clean, see Fig. 7 [37]. The shark skin riblets are believed to increase water flow rate at the solid–liquid interface, which effectively lowers drag and prevents the buildup of biofouling. The microstructured riblets encourage anisotropic flow as well as control vortices on the skin naturally present in turbulent flow. The riblets lift and presumably pin any vortices generated in the viscous sublayer, which leads to lower drag and antifouling. Lifting reduces the total shear stress since vortices contact just the small riblet tips (as opposed to the total surface area). Pinning is believed to reduce the cross-stream motion of fluid (i.e. provides anisotropic flow) and ejection of vortices from the viscous sublayer, which

reduces momentum transfer [28–30,35,37]. Turbulent vortices are also present in fluids besides water; therefore shark skin inspired riblets provide drag-reducing benefit with other fluids such as air and oil [37].

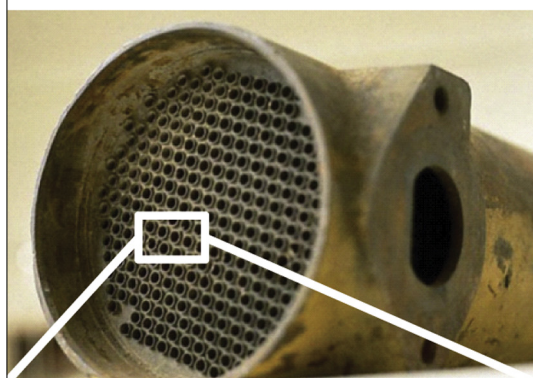
Lotus leaf hierarchical micropapillae, which are microbumps superimposed with low surface energy waxy nanostructures, provide superior self-cleaning properties [38,81]. It should be noted that several other plant species also exhibit self-cleaning properties due to superhydrophobic and low adhesion characteristics. In nature, rain droplets impact the lotus leaf surface and effectively roll off due to the water repellency effect. This leads to self-cleaning, where droplets collect and remove any contaminant particles. The lotus effect relies on roughness induced superhydrophobicity and low adhesion, where air pockets are present at the solid–liquid interface [2–4,38,40].

Antifouling surfaces have been the subject of many studies using shark skin inspired riblet surfaces. An ideal surface should withstand harsh environments, adhere to a variety of substrates, combine both low drag and antifouling properties, and be relatively inexpensive. Shark skin inspired products include the drag-reducing 3 M Corp. (Minneapolis, MN) experimental vinyl riblets [82] as well as the Speedo FastSkin® fabric racing swimsuit. In an effort to create antifouling surfaces, the shark skin inspired Sharklet™ microtextured pattern has been developed and reportedly reduces biofouling. This microtextured film with riblet inspired morphology operates on the principle that microorganisms are deterred from surfaces with crevices slightly smaller than themselves [31,58,59,61,75].

3.2. Rice and butterfly wing effect

Bixler and Bhushan [63] reported that individual biogenic structures such as rice leaves and butterfly wings exploit a combination of antifouling properties, such as the shark skin and lotus leaf effects. It is believed that since rice plants thrive in humid, marshy environments, self-cleaning prevents unwanted fouling, which may inhibit photosynthesis [63]. Since butterflies are fragile and

Inorganic fouling examples



Clean (top) fouled (bottom) power plant heat exchanger



Heat exchanger tubing with crystallization

Fig. 4. Inorganic fouling examples [7]. Inside of a clean nuclear power plant heat exchanger tube (top) compared to a fouled tube with corrosion deposits (middle) [24]. Calcium carbonate crystallization fouling on the outside of a heat exchanger (bottom) (photo courtesy of H&C Heat Transfer Solutions, Inc.).

unable to clean their wings, these properties are critical to maintain structural coloration and flight control [83]. Both rice leaves and butterfly wings exhibit unique anisotropic flow characteristics, where droplets roll down the blade of the rice leaf and axially away from the butterfly body [63].

Shown in Fig. 8a are the hierarchical structures of rice leaves that provide the anisotropic flow, superhydrophobicity, and low adhesion properties. Although this study focuses on rice leaves (as explained in later sections), the butterfly wing mechanisms are also described for a more compressive understanding. For rice leaves and butterfly wings, anisotropic flow is facilitated by the longitudinal grooves with a transverse sinusoidal pattern in rice leaves and aligned shingle-like scales in butterfly wings [63].

Superhydrophobicity and low adhesion is facilitated by micropapillae superimposed by epicuticular waxy nanobumps in rice leaves [63,84–87] and hierarchical scales with microgrooves on butterfly wings [63,83,88–90].

The combination of anisotropic flow, superhydrophobicity, and low adhesion is reported to facilitate self-cleaning [63,64]. A simplified conceptual model of rice leaf hierarchical structures is shown in Fig. 8b. The water droplets sit above the hierarchical surface structures and can effortlessly roll and collect contaminants to improve self-cleaning efficiency [63,64].

4. Antifouling: current prevention and cleaning techniques

Bixler and Bhushan [7] report that reducing or eliminating fouling is accomplished through a variety of methods in medical, marine, and industrial applications. Common controls exist, such as microorganism resistant coatings and self-cleaning low adhesion surfaces [7]. Improving biocompatibility can also reduce medical biofouling, since microorganisms will be less likely to target any implanted device [91–93].

4.1. Preventative techniques

Medical biofouling is of great concern for the health of patients. Fouling of medical devices can significantly alter the properties of the fouled materials. Multiple coatings and anti-fouling techniques have been reported [94], and many incorporate hydrophobic polymers with antimicrobial properties [7,95]. Example applications include catheters, endotracheal tubes [96], and orthopedic hip implants. Superhydrophobic and low adhesion “self-cleaning” surfaces are believed to minimize protein bioadhesion, and antimicrobial surfaces reduce ambient microorganisms [65]. Surfaces with superhydrophilic polyethylene glycol or polyethylene oxide exhibit less protein adsorption and bioadhesion [97,98]. Other coating examples include self-assembled monolayers (SAMs), which are desirable since they coat surfaces with a molecularly thick biocompatible film. This is especially important for applications such as biomedical micro/nanoelectromechanical systems (BioMEMS/NEMS) [91–93,99]. Fig. 9 shows an orthopedic hip implant covered with a biocompatible coating designed to control biofouling, wear rate, and promote bone growth [100,101].

Additional medical preventative techniques include hydrophilic red blood cell plasma membranes that reduce biofouling in medical devices and other synthetic membranes [102]. For instance, Advanced Hydro[®] hydrophilic polymer coatings on water purification and filtration membranes provide a 30–50% reduction in fouling (<http://www.advancedhydro.net>). Clean Membranes[®] utilize a “smart-comb” copolymer in ultrafiltration membranes to provide antifouling and increased permeability (<http://www.cleanmembranes.com>). Superhydrophilic and antimicrobial nanoparticles in a polyamide thin film coating provide antifouling for reverse osmosis (RO) membranes (<http://www1.cnsi.ucla.edu>). Furthermore, studies show success using titanium dioxide (TiO₂) nanoparticles as a thin film on reverse osmosis membranes to control biofouling due to the photocatalytic effects that reduce bacteria [103]. Similarly, ultrafiltration membranes impregnated with antimicrobial and hydrophilic silver nanoparticles (nAg) provide antifouling benefit. Fig. 9 compares such treated and untreated polysulfone ultrafiltration membranes, showing fewer *E. coli* (*Escherichia coli*) attached to the nAg membrane [104].

Tributyltin coatings have been efficacious in marine antifouling application, and were commonplace. Unfortunately, toxicity and other properties make tributyltin coatings unsuitable for in vivo use. Because of tributyltin toxicity to marine organisms, antifouling preparations containing it were banned in 2008 by the

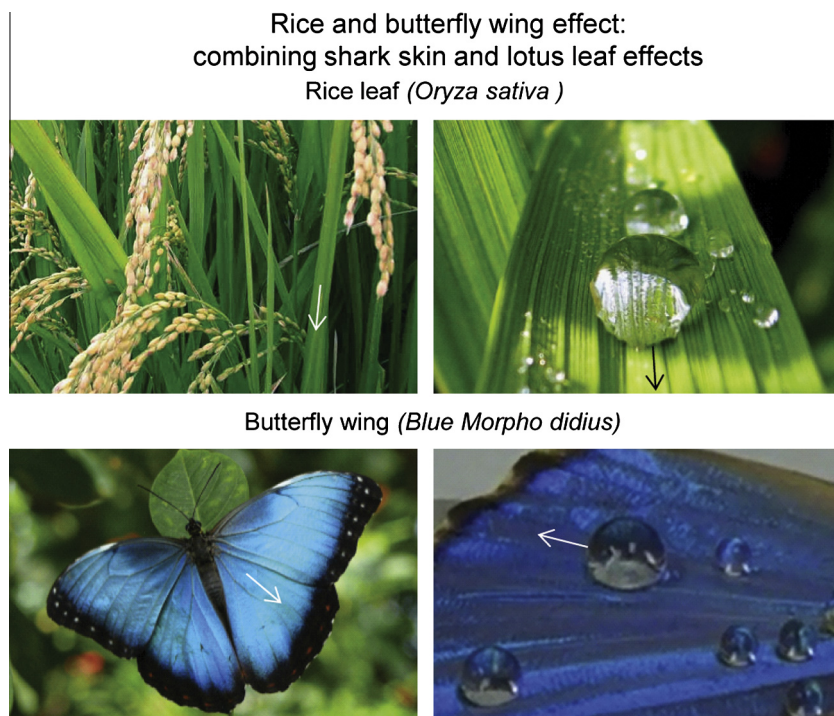


Fig. 5. Rice and butterfly wing effect that combines the shark skin and lotus effects [48]. Shown are water droplets resting atop their superhydrophobic and low adhesion surfaces. Rice leaves contain longitudinal grooves with a transverse sinusoidal pattern and butterfly wings contain aligned shingle-like scales that provide anisotropic flow. Hierarchical structures consisting of micropapillae superimposed by waxy nanobumps in rice leaves and microgrooves on shingle-like scale structures in butterfly wings provide superhydrophobicity and low adhesion. This combination of anisotropic flow, superhydrophobicity, and low adhesion is believed to lead to improved antifouling and self-cleaning [25,48,63].

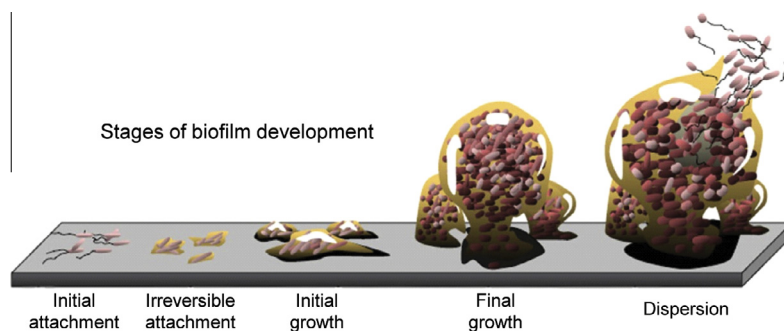


Fig. 6. Biofilm development process. Schematic illustrates the five stage colonization process (adapted from D. Davis).

International Maritime Organization, adding urgency to the development of more environmentally friendly antifouling and foul-release coatings. As a result, many researchers have been exploring environmentally friendly methods such as new nontoxic antifouling paints and foul-release coatings [13,15,16,105]. Marine foul-release coatings are nontoxic hydrophobic surfaces that promote self-cleaning. Nano/microtextured surfaces control surface energy, charge, conductivity, porosity, roughness, wettability, and friction [106]. Hydrophobic materials such as fluoropolymers and silicone elastomers are nontoxic, adhesion resistant, environmentally friendly, and easy to clean [107]. Siloxane elastomers provide a combination of lower elastic modulus and lower surface energy, creating a hydrophobic, low-stick surface. Silicone foul-release coatings are used for some quick moving boats, but wear away over time [98,108]. Fig. 9 shows a foul-release organic tin-free hull coating called “Green Ocean Coating Heavy Duty” by Advanced Marine Coatings, whose smoothness and hardness reduces cleaning cycles (<http://www.bayermaterials.com>).

Industrial antifouling is accomplished similarly to marine antifouling, with coatings that prevent and remove fouling. For instance, industrial heat exchanger applications utilize SAMs to reduce bioadhesive strength between foulers and a heated surface, using low surface energy hydrophobic properties. Materials such as silicon can withstand temperatures up to 200 °C whilst materials such as hexadecyl disulfide can withstand up to 225 °C. SAMs furthermore offer corrosion protection by limiting the oxygen and water diffusion at the substrate surface. This technique could be widely used for heat exchangers but needs long term durability testing [109].

4.2. Self-cleaning surfaces and cleaning techniques

Even though fouling prevention is the goal, in many cases fouling is not completely preventable and a surface must be cleaned. In response, several self-cleaning coatings have been developed for various applications. Bixler and Bhushan [7] report example self-

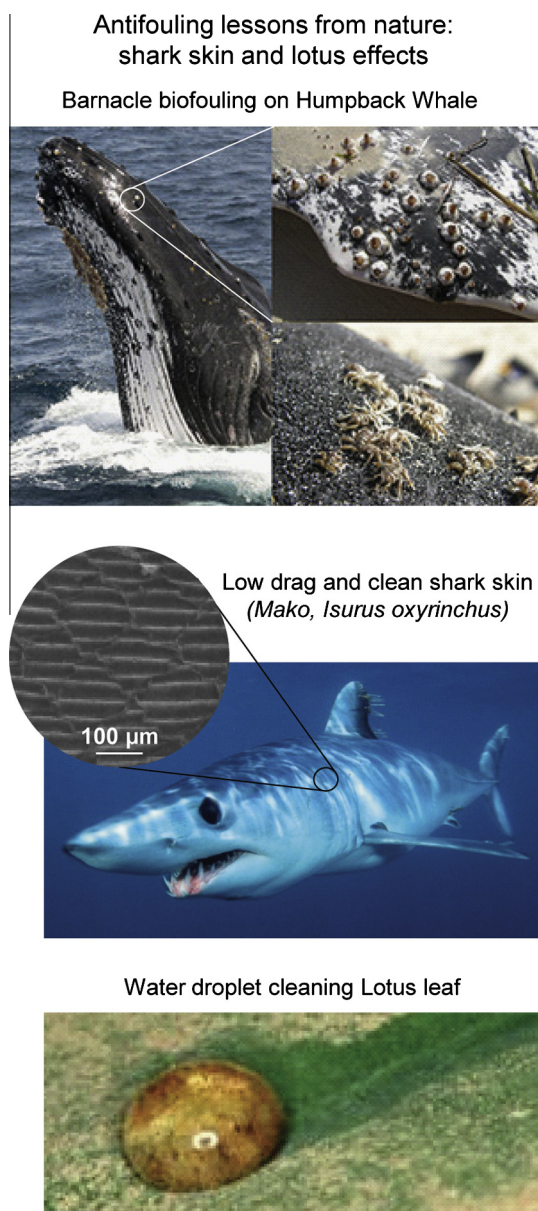


Fig. 7. Shark skin and lotus leaf effects. Shown is biofouling on the whale and clean shark skin, even though both live in similar environments. The SEM image shows Mako shark skin microstructured riblets that provide low drag and antifouling properties [37]. The lotus effect is illustrated with a water droplet carrying off contaminants from the lotus leaf surface [7].

cleaning products including hydrophobic paints, roof tiles, and fabrics as well as self-cleaning hydrophilic windows and membranes. Examples are Lotusan® paints by Sto (<http://www.stocorp.com>), Tegotop® paints by Degussa (<http://www.goldschmidt.com>), Erlus Lotus® roof tiles (<http://www.erlus.com>), NanoSphere® fabrics by Schoeller Technologies (<http://www.schoeller-textiles.com>), and SunClean® glass by PPG Industries (<http://www.ppg.com>). Self-cleaning is demonstrated with the hydrophobic Lotusan® roll on and Tegotop® spray paints that create a microstructure surface (<http://www.stocorp.com>; <http://www.goldschmidt.com>), hydrophobic Erlus Lotus® roof tiles with an etched surface finish (<http://www.erlus.com>), and hydrophobic NanoSphere® fabrics with integrated nanoparticles (<http://www.schoeller-textiles.com>). Self-cleaning hydrophilic SunClean® glass with titanium dioxide uses the sheeting action of water to efficiently gather and remove contaminants (<http://www.ppg.com>). Fig. 10 shows the

improved cleanliness with Lotusan® hydrophobic self-cleaning paints on concrete and stucco (<http://www.stocorp.com>), and better clarity with the SunClean™ hydrophilic self-cleaning glass (<http://www.ppg.com>).

Physical cleaning and chemical disinfecting are often considered effective methods to remove fouling [7,12,14,16,110,111]. Most employ mechanical methods such as manual scrubbing to disrupt and remove the fouling. Fig. 11 illustrates common methods of cleaning and disinfecting such as medical sterilization, chemical disinfection, high-pressure water spray, automatic brush scrubbing, and membrane cleaning techniques. Included are a steam sterilization medical autoclave, Listerine® antiseptic mouth rinse, water spray on a low-adhesion ship hull, automatic brush washing system, and microfiltration membrane using sonication, chemicals, and water backwash [7,112]. Disinfecting is commonly employed in dental applications with antiseptic mouth rinses such as Listerine®, for example, the active ingredient chlorhexidine attacks and reduces dental biofilm to control plaque and gingivitis in the oral cavity [113]. The biofilm is reduced when the individual bacteria walls rupture (<http://www.jjidentalprofessional.com>). Mouth rinses combined with brushing and flossing increase the effectiveness, since such “mechanical methods” disrupt the biofilm [111].

5. Experimental methods

In this section, we explain the origin of the new rice leaf inspired surfaces, sample fabrication processes, and experimental apparatuses which provide antifouling, self-cleaning, contact angle, and contact angle hysteresis measurements.

5.1. Bioinspired rice leaf surfaces

By studying nature’s secrets to develop antifouling and self-cleaning surfaces, the rice leaf is particularly intriguing due to its relatively simple micropapillae morphology [48,63,64]. For instance, bioinspired rice leaf surfaces using micro-sized pillar patterns can be produced with silicon master patterns fabricated by photolithography [48]. Sample geometrical dimensions are based on actual rice leaf morphology from a previous study [63], and are shown in Table 2. Actual rice leaf surfaces are covered by a sinusoidal pattern of micropapillae with height 2–4 μm, diameter 2–4 μm, pitch 5–10 μm, and peak radius 0.5–1 μm [63]. Bioinspired rice leaf surfaces shown in Fig. 12 were created to study the rice and butterfly wing effect, and were designed to include anisotropic flow and superhydrophobic characteristics.

Samples with geometrical dimensions for the current study are based on samples from a previous study [48], and are shown in Table 3 with pillar height, pillar diameter or rib width, and pitch. Samples 1 and 2 represent just the pillars found in actual rice leaf patterns with pillar heights of 2 and 4 μm, respectively with a pillar diameter of 5 μm and pitch of 7 μm. Sample 3 represents the bioinspired rice leaf surface with dual height pillars selected to create a sinusoidal pattern whereas in Sample 4, the surface is created with a combination of 2 μm high pillars and 4 μm high ribs. Dual height features in Samples 3 and 4 create a hierarchical structure that is believed to facilitate the anisotropic flow similar to the sinusoidal grooves in actual rice leaves. Sample 3 most closely simulates rice leaf micropapillae morphology whereas Sample 4 includes pillars plus ribs to further enhance anisotropic flow. These designs were developed based on available microfabrication techniques to produce master patterns in silicon wafers.

Pitch length of the microstructured features found in Samples 1–4 resembles that of actual rice leaves, in order to promote antifouling [64]. Averaging the actual rice leaf micropapillae pitch

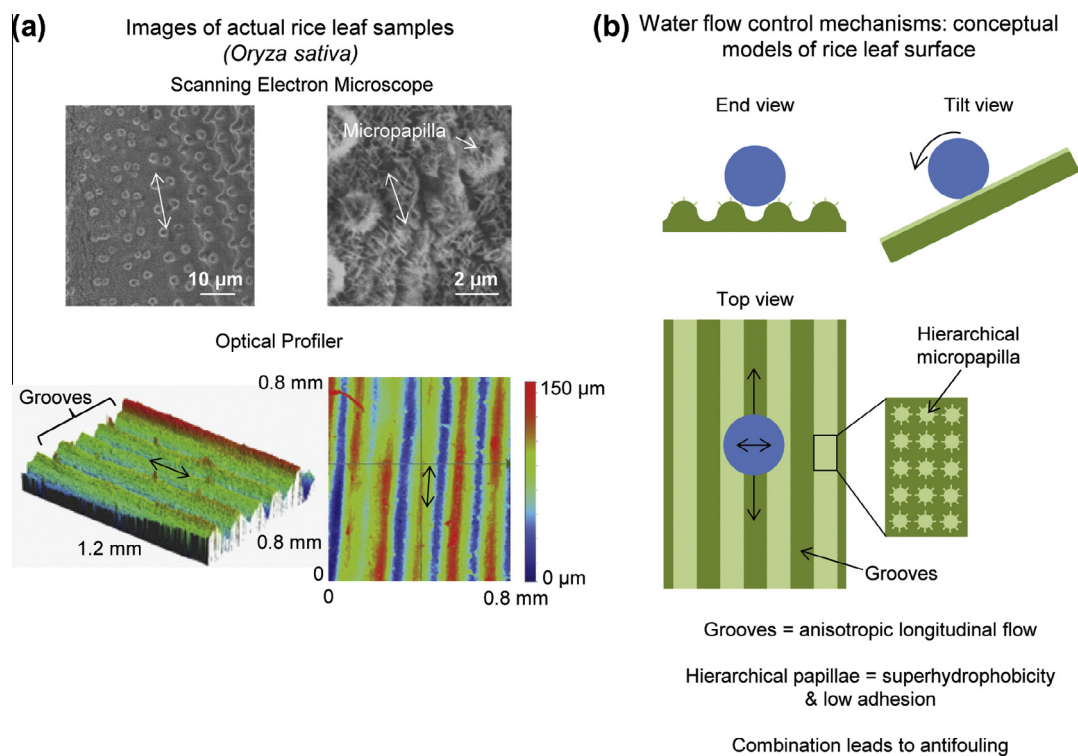


Fig. 8. Optical profiler height maps of (a) actual rice leaf as well as (b) a conceptual model of water flow control mechanisms (adapted from Bixler and Bhushan [63]). Arrows indicate the tendencies of fluid flow in transverse and longitudinal directions. These provide necessary information for surface morphology characterization.

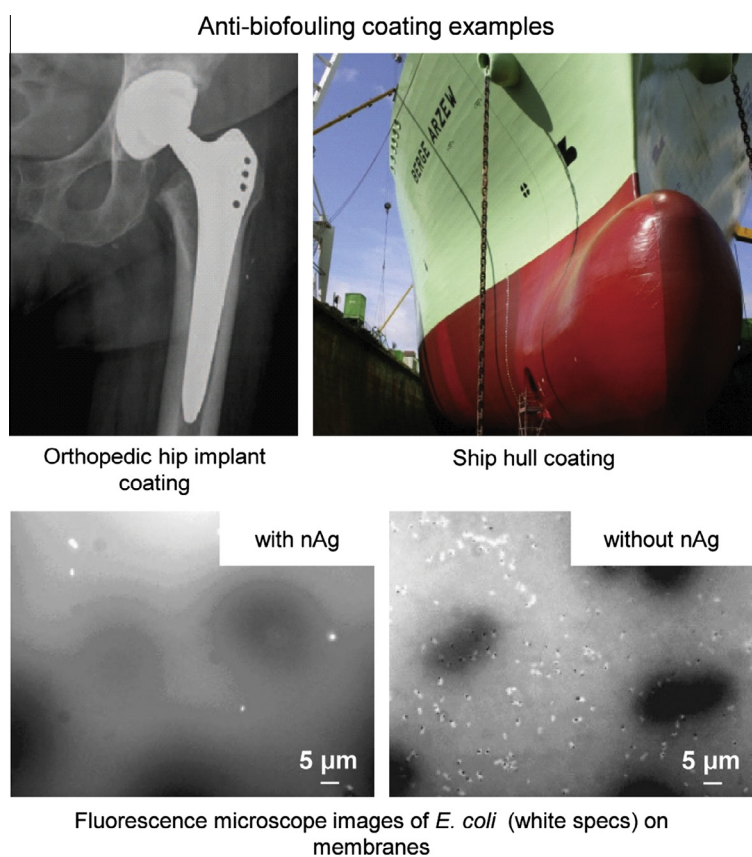


Fig. 9. Medical, marine, and industrial antifouling coating examples [7]. Orthopedic hip implants are covered with biocompatible coatings designed to reduce biofouling, wear rate, and promote bone growth (top left) (photo courtesy of Ir Jef Vleugels). Marine foul-release organic tin-free hull coating called “Green Ocean Coating Heavy Duty” minimizes manual cleaning (top right) (photo courtesy of Bayer MaterialScience). Polysulfone ultrafiltration membrane treated with silver nanoparticles (nAg) shows antifouling benefit with fewer *E. coli* attached (bottom) [104].

Anti-inorganic fouling coatings

Lotus effect self-cleaning paint



Conventional paint

StoCoat™ Lotusan®

Hydrophilic self-cleaning windows



Conventional glass

SunClean™ glass

Fig. 10. Effectiveness of anti-inorganic fouling coatings [7]. Demonstrated is the improved cleanliness with Lotusan® hydrophobic self-cleaning paint (top) (<http://www.stocorp.com>), and improved clarity with the SunClean™ hydrophilic self-cleaning glass (bottom) (<http://www.ppg.com>).

length yields the pitch length of 7 μm for Samples 1–4. It is believed that antifouling is most efficient when the gap between features is less than the approximate diameter of fouling microorganisms [31,58,59,61,75].

5.2. Silicon master patterns

Fabricating bioinspired rice leaf surfaces with micro-sized features required following previously demonstrated microfabrication techniques in order to achieve the desired dimensions and tolerances [48]. As such, Samples 1–4 were created using silicon master patterns (fabricated by TRICR Corp., San Francisco, CA) and soft-lithography techniques. Solid models of Samples 1–4 were designed with computer aided design software by Solidworks® (Dassault Systèmes SolidWorks Corporation, Waltham, MA). A single square photomask (130 mm $\times$ 130 mm $\times$ 2.3 mm thick with the critical dimensional tolerance of $\pm 0.025 \mu\text{m}$) was produced containing each pattern for Samples 1–4. The photomask was soda lime with low reflectance chrome plating showing pillars represented as transparent features on a solid background. Silicon master patterns for Samples 1–4 were created on separate 4" (10.16 cm) round silicon wafers with sample areas measuring 6 mm by 100 mm. The etching procedure included standard wafer dehydration, wafer prime, resist coating, wafer exposure, post exposure bake, wafer development, and hard bake. Tolerances include a 10% positional feature and ± 0.25 micron pit depth tolerance.

Silicon master patterns with pits (negative versus positive features) were fabricated in order to emboss polymer films and to accurately produce master patterns [48]. Embossing protrusions onto polymer films using master patterns is possible when the

master pattern contains pits. In addition, pits are less challenging to fabricate as compared to protrusions for the dual height features of Samples 3 and 4. For instance, to create protrusions with two heights, all protrusions would need etched to the taller height using one photomask pattern, and then the shorter protrusions would need etched again with a second photomask pattern. Difficulty arises due to challenges of aligning the second photomask pattern to further etch to the shorter heights. Tolerances exist in both the photomask and alignment procedures which significantly increase the possibility of defects. This is not the case when etching two pit depths, which does not require the alignment of the second photomask pattern on top of existing protrusions. Instead, each pit depth is created with the same photomask pattern realigned and with different etching times [48].

5.3. Urethane soft lithography procedure

Rice leaf replica surfaces with hierarchical features were previously fabricated using a two-step soft lithography procedure and a nanostructured coating [63]. The soft lithography procedure included first fabricating a negative mold in silicone and then positive replicas in urethane. A negative mold was created with liquid platinum silicone (Smooth-On Dragon Skin 20) poured onto actual rice leaves that were cleaned with purified water and isopropyl alcohol. The liquid silicone ensured that details were accurately replicated and that air bubbles would rise away from the molding surface. With the silicone mold complete, liquid urethane polymer (Smooth-On Smooth-Cast 305) was applied and cured, yielding a precise positive replica [63].

For the nanostructured superhydrophobic coating, silica particles were selected as they are known to provide high durability and transparency, if desired [114,115]. Rice leaf replicas were dip-coated with an ultrasonic mixed solution consisting of 50 nm (± 15 nm) hydrophobized silica nanoparticles (by Evonik-Degussa Corporation, Parsippany, New Jersey) combined with methylphenyl silicone resin (SR355S from Momentive) dissolved in tetrahydrofuran and isopropyl alcohol [63].

5.4. PDMS soft lithography procedure

Rice leaf bioinspired surfaces with micro-sized features were fabricated following a previously demonstrated three-step soft lithography procedure [48]. For Samples 1–4, this procedure was utilized in order to produce samples from the silicon master patterns. The material polydimethylsiloxane (PDMS) (184 Silicone Elastomer by Dow Corning) was chosen due to its low surface energy property which leads to high contact angle. Positive master molds were fabricated with hydrophilic vinyl polysiloxane impression material (Examix™ Type 3 low viscosity by GC America Inc.), which successfully de-molded from the silicon [48]. The silicon master patterns were cleaned using acetone and isopropyl alcohol rinses with nitrogen drying. The impression material was deposited directly onto the silicon master patterns via cartridge gun and mixing nozzle, gently pressed with a flat block, and cured at room temperature for 4 h.

The positive molds in dental impression material were used to create negative molds in urethane material. Liquid urethane polymer (Smooth-On Smooth-Cast 305) was chosen due to its dimensional stability, ease of casting, ability to de-mold easily from the dental impression material, and ability to de-mold from PDMS in the final step. The liquid urethane was applied and cured, yielding precise negative master molds. Finally, the last step entailed utilizing the negative molds in urethane to create final samples in PDMS. Liquid PDMS was manually mixed, poured onto the urethane negative master molds, and allowed to cure at room temperature for 48 h. The liquid PDMS remained in the liquid-form for at least

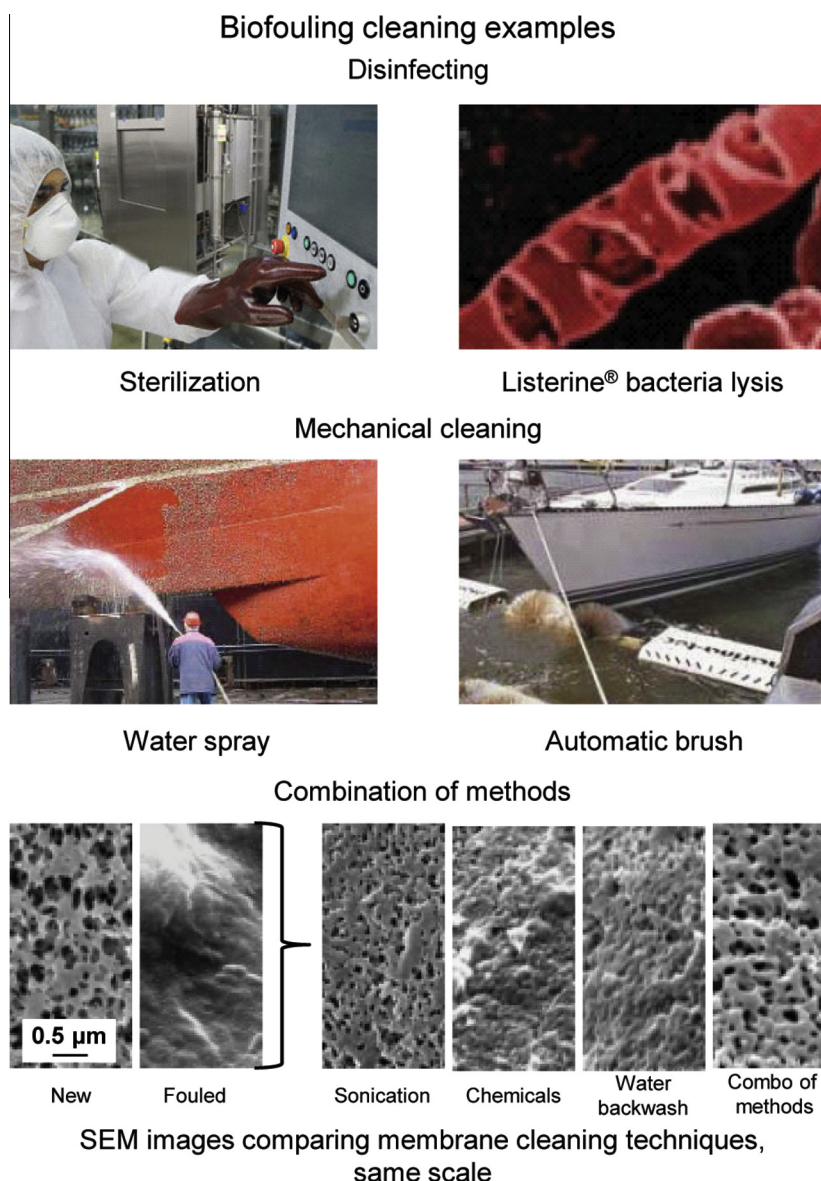


Fig. 11. Medical, marine, and industrial biofouling cleaning examples [7]. Steam sterilization medical autoclave (top left) (photo courtesy of Life Science Outsourcing), Listerine® effect on bacteria (top right) (adapted from Christoph Schaudinn), water spray on a low-stick ship hull (middle left), automatic brush washing system (middle right) [122], and microfiltration membrane using sonication, chemicals, and water backwash (bottom) (adapted from Lim and Bai [112]).

Table 2

Physical characterization of surface structures from actual rice leaves (adapted from Bixler and Bhushan [63]).

Sample	Actual						
	Description		Vertical –dim (μm)	Transverse–dim (μm)	Longitudinal–dim (μm)	Pitch (μm)	Peak radius (μm)
Rice leaf (<i>Oryza sativa</i>)	Sinusoidal grooves array covered with micropapilla and nanobumps	Grooves Micropapillae	125–150 2–4	150–175 2–4 dia	Full length n/a	n/a 5–10	n/a 0.5–1

12 h of the 48 h duration, therefore allowing any air bubbles to effectively escape. The final samples in PDMS readily de-molded from the urethane negative master molds. This step was repeated several times in order to produce the necessary quantity of samples in PDMS.

5.5. Hot embossing procedure

Rice leaf bioinspired surfaces with micro-sized features were fabricated following a previously demonstrated hot embossing

procedure [48]. Embossing polymer films was conducted using a commercial hot embossing machine (model EVG 520, S/N S050007, software V.7.01.00.00) in a clean room environment at the Nanotech West facility (The Ohio State University, Columbus, OH). Sample 3 was selected to be embossed for further evaluation due to its superior self-cleaning properties in PDMS (see comprehensive rice leaf inspired surface results in Bixler and Bhushan [48,64]). The hot embosser produced single-sided embossed polymer films from silicon master patterns using a combination of heat and pressure. The polymer films were selected to be optically clear

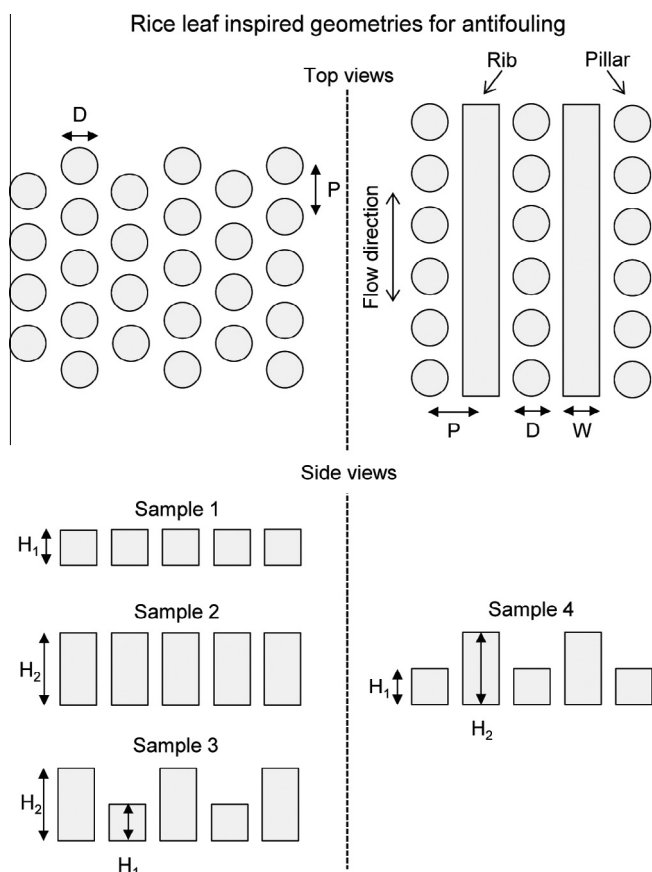


Fig. 12. Rice leaf inspired geometries for antifouling [48]. Shown are microstructured surfaces with pillars and ribs that provide discontinuities with sharp edges that discourage fouling. Pillars in Samples 1–3 are arranged in a hexagonal array in order to maintain consistent gaps to promote antifouling. Sample 3 includes dual height pillars whilst Sample 4 combines pillars and ribs.

175 μm thick acrylic film based upon PMMA and acrylic elastomers (Evonik Cyro ROHAGLAS® 99524). PMMA molds relatively easily due to its low softening temperature point and the 175 μm thickness was deemed sufficient for embossing 4 μm tall features. The sandwich assembly consisted of a 4" (10.16 cm) round silicon master pattern on the bottom, 4" (10.16 cm) round PMMA sheet in the middle, and 4" (10.16 cm) round flat silicon wafer on the top. The PMMA film and silicon wafers were thoroughly cleaned using acetone and isopropyl alcohol rinses with nitrogen drying prior to assembly [48].

The PMMA film required a low surface energy coating on both sides for proper de-molding from the silicon master patterns and flat silicon wafers. The aqueous fluorochemical Capstone™ ST-100 (Dupont) was selected and diluted to 12% by volume with deionized water and mixed using an ultrasonic mixer. A spray deposition method was chosen for a uniform thin coating by using an internal mixing double action airbrush (model VL-SET, Paasche) and laboratory air at 30 psi [48].

The hot embossing procedure was developed from previous studies [46] as well as trial and error by Bixler and Bhushan [48]

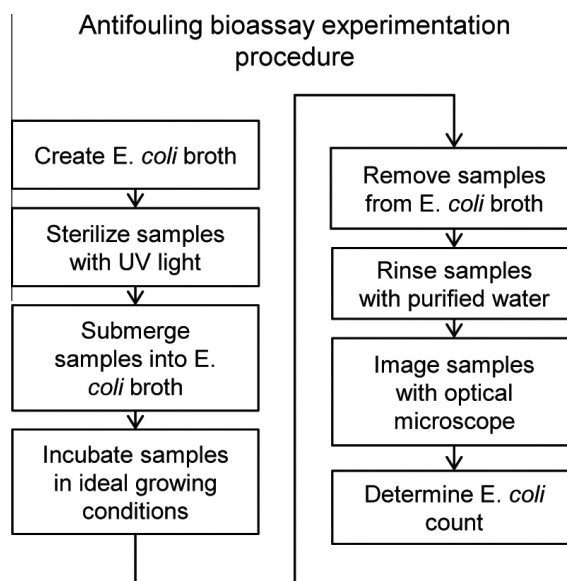


Fig. 13. Antifouling bioassay experimental procedure with *E. coli* cells in the laboratory. As shown, samples are sterilized, submerged into broth, placed in incubator, rinsed to remove loose cells, and imaged with oil immersion light microscope to determine cell counts.

by investigating suitable pressures, temperatures, hold times, release coatings, and PMMA thickness. Once assembled, the hot embosser chamber was vacuumed to 40 Torr (50 mbar) and the top and bottom chucks heated in order to soften the material. The PMMA has a Vicat softening point of 100 °C (Cyro Industries), therefore chucks were heated to 150 °C to ensure proper polymer softening. Once heated, an axial clamping force was applied and increased in increments of 2000–10,000 N with a hold time of 15 s at each step. At the maximum load, the clamping force was maintained for 10 min. Upon completion, the sandwich assembly was cooled to 80 °C and then the clamping force was released and the chamber vented to atmospheric pressure. De-molding was possible using a razor blade around the edges without fracturing the silicon master patterns [48].

5.6. Anti-biofouling measurements

In order to study the anti-biofouling benefit of the various samples, several foulers were considered primarily based on their size and availability. These included zoospores (*Enteromorpha*) [56] and (*Ulva linza*) [116], proteins [117], *E. coli* [118], and MRSA (*Methicillin-resistant Staphylococcus aureus*) [119]. The microorganism *E. coli* was selected due to its prevalence in the medical industry, availability, and access to facilities for safe handling. The multi-step anti-biofouling measurement process is shown in Fig. 13 with each process from the *E. coli* broth creation to the final counting. The dimensions of single wild type *E. coli* cells are approximately 0.5–0.75 μm wide and 2 μm long. When actively growing and dividing, the cells stretch to $\sim 4 \mu\text{m}$ before dividing into two cells.

The bioassay procedure began with stock solution of *E. coli* (HB2151 *E. coli* clones from The Medical Research Council,

Table 3
Geometric dimensions of rice leaf inspired samples (Bixler and Bhushan [48]).

	Description	Height (H_1 or H_2) (μm)	Pillar diameter (D) or rib width (W) (μm)	Pitch (P) (μm)
Sample 1	Hexagonal array of single height pillars	2	5	7
Sample 2		4		
Sample 3	Alternating rows of dual height pillars	2 and 4		
Sample 4	Alternating rows of single height pillars and ribs	2 (pillars) 4 (ribs)		

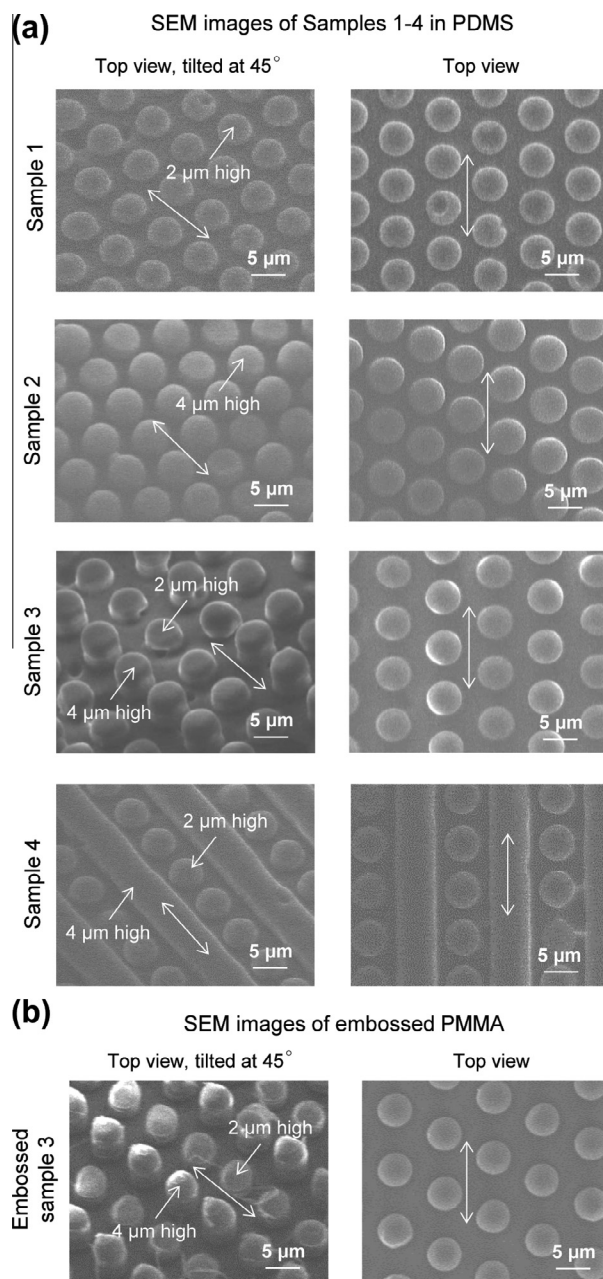


Fig. 14. Montage of SEM images for Samples 1–4 and Embossed sample 3 [48]. Arrows indicate direction of droplet movement for self-cleaning experiments. Each sample is shown with the top view tilted at 45° (left side column) and with no tilt (right side column) in order to highlight the feature heights. (a) Samples 1 and 2 images show the uniform height pillars arranged in the hexagonal array. Sample 3 images show the dual height pillars and Sample 4 images show the combination of pillars and ribs. Each sample was successfully fabricated in PDMS using the prescribed three-step soft lithography procedure. (b) Embossed sample 3 shown indicates that the soft-lithography in (a) and Embossed sample 3 exhibits similar feature characteristics.

Cambridge, England) streaked onto TYE (Tryptone and Yeast Extract) dehydrated agar plates for bacterial colony growth. The plates were then placed in an incubator (Labnet, model 311 DS) at 37 °C overnight, and then stored at 4 °C until use. To prepare *E. coli* broth, a single colony was picked from streak plates and inoculated into 50 ml sterile LB (Luria Broth, Lennox from Fisher Scientific, Pittsburgh, PA) in a beveled flask. *E. coli* was grown for 16 h with shaking at 250 rpm at a temperature of 37 °C. After incubation, *E. coli* solution was diluted to 1/100 and 1/10 into sterile LB broth and placed on ice until use.

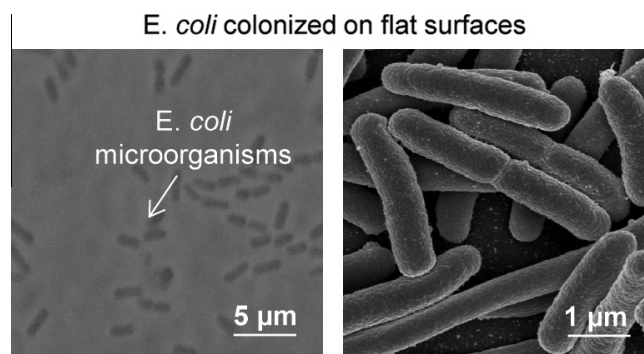


Fig. 15. Images of *E. coli* on flat surfaces after the bioassay procedure (present work) and at a higher magnification (adapted from <http://www.universityofcalifornia.edu>).

The flat control, Samples 1–4, and Embossed sample 3 were placed in the bottom of a sterile, polystyrene culture dish and sterilized for 10 min via ultraviolet exposure (30 W T8 Hot Cathode Germicidal Sterilamp fluorescent light bulb, Philips TUV30 W G30T8 360164). *E. coli* broth was added to the culture dish until all samples were covered and then incubated at 37 °C for either 2 or 4 h. The aforementioned concentrations and incubation times were determined from trial and error with samples to achieve an appropriate level of *E. coli* growth. The goal was to visually determine a difference at 1000× magnification between flat control and other samples under investigation.

After incubation, samples were removed from *E. coli* broth and rinsed at a 45° angle. Rinsing removed any unattached *E. coli* from the surface prior to imaging. An auto-pipette (Pipet-Aid from Drummond Scientific, Broomall, PA) was clamped and held on a stand over the rinsing station with a liquid fall height (distance between pipette tip end and sample) of 7 cm. Samples were rinsed with 10 ml of sterile PBS at a flow rate of approximately 2 ml/s, and then immediately placed onto a glass microscope slide with a glass cover slip on top. Using a light microscope and CCD camera (Nikon, Diagnostics Instruments, Inc.), a 140 μm by 140 μm area of each sample was imaged and flat samples analyzed with image processing software (SPIP 5.1.11, Image Metrology A/S, Horsholm, Denmark). Images were collected with a 100× oil immersion objective lens for a total magnification of 1000×. For flat control samples, image processing software (SPIP 5.1.11, Image Metrology A/S, Horsholm, Denmark) was utilized to determine total coverage area of *E. coli*. For Samples 1–4 and Embossed sample 3, the *E. coli* was manually counted five times in circular areas 35 μm in diameter using two samples within the same bioassay.

5.7. Anti-inorganic fouling measurements

To understand the anti-inorganic fouling effectiveness of the samples, self-cleaning experiments were previously conducted [48]. These experiments included contaminating samples, employing a wash technique, and determining the percentage of inorganic particles removed. Depositing contaminate particles on the tilted (45°) replicas involved a glass contamination chamber (0.3 m in diameter and 0.6 m high). A tray containing 0.2 g of hydrophilic silicon carbide (SiC) contaminants (400 mesh particle size by Aldrich, with sizes ranging from 10 to 15 μm) was placed in the top chamber with an air hose directed in the center. These particles were chosen because of their similar properties to natural dirt (shape, size, and hydrophilicity). Contaminants were blown with laboratory air for 10 s at 300 kPa, and then allowed to settle for 30 s before the separator panel was removed.

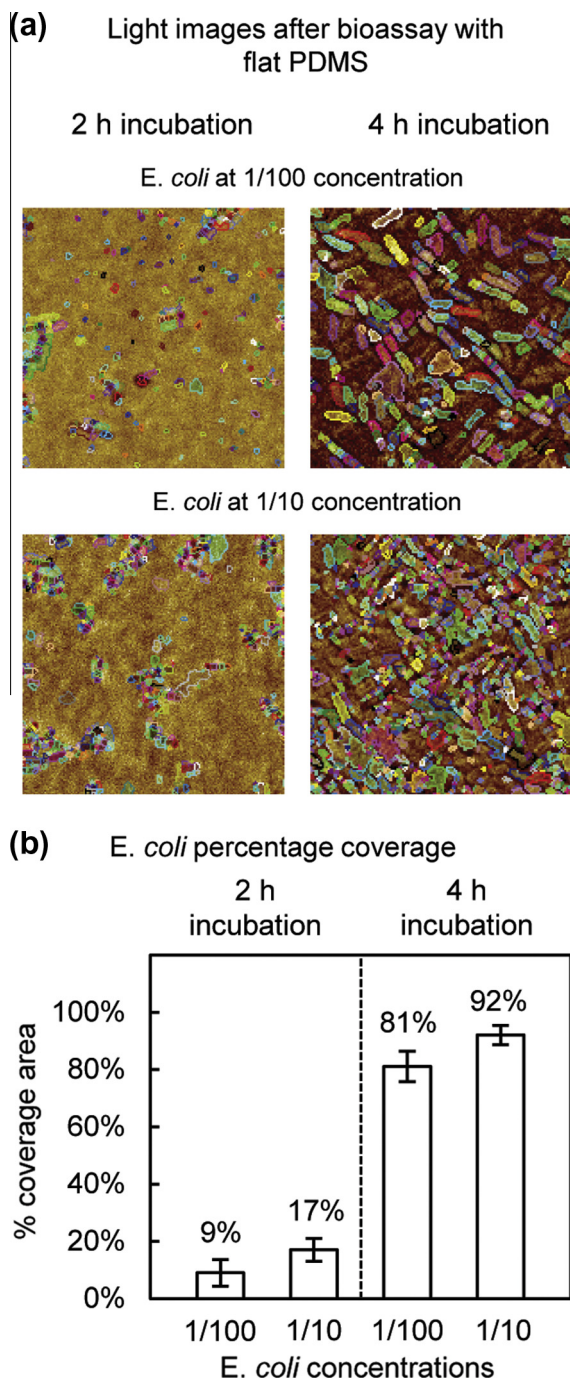


Fig. 16. (a) Light microscope images of flat PDMS control samples after *E. coli* bioassay with two concentrations and two incubation times. (b) As indicated, *E. coli* covers the majority of the surface after 4 h in ideal growing conditions. This baseline serves to compare the various Samples 1–4 and Embossed sample 3 when calculating percentage reduced coverage.

After 30 min, the samples were removed and subjected to pre-wash experiment particle analysis. Using a light microscope and CCD camera (Nikon, Optihot-2), a 280 μm by 210 μm area of each sample was imaged and analyzed with SPIP image processing software. This process was performed before and after each wash experiment. Wash experiments consisted of exposing the tilted (45°) replicas to water droplets falling from a specified heights and drip rates (total duration of 2 min using 10 mL water). Droplet velocities were approximately 1 and 5.6 m/s at heights of 0.02 and 0.4 m, respectively. This translates into pressures of 200 and 4000 Pa, respectively.

5.8. Wettability measurements

To understand the effects of wettability on antifouling, apparent contact angle (CA) and contact angle hysteresis (CAH) experiments were previously conducted [48]. CAH is the difference between the advancing (downhill side) and receding (uphill side) contact angles, which is lower for Cassie-Baxter (droplet sitting on top of asperities) and higher for Wenzel (droplet penetrating gaps between asperities) regimes. High CAH and adhesion strength explains the deformed shape of a water droplet adhered to a vertical window [40]. Purified water droplets approximately 1 mm in diameter (5 μL) were deposited and measured with an automated goniometer (Rame-Hart model 290-F4). CAH was determined by tilting the sample until the droplet began to move (up to 90°), and subtracting the advancing and receding contact angles.

6. Results and discussion

In this section, we present images and measurements characterizing the various samples under investigation. Presented are measurements of various features describing surface morphology and replication success as well as results from antifouling, self-cleaning, and wettability experiments. Conceptual models are shown to provide an understanding of mechanisms at work for the rice and butterfly wing effect samples.

6.1. Sample characterization

To characterize the geometric dimensions of the various samples, SEM images previously collected by Bixler and Bhushan [48] provide a qualitative and quantitative comparison and understanding of the relevant parameters. Images for each Samples 1–4 are shown in Fig. 14a, with top views tilted at 45° and not tilted. Arrows on each image provide the flow direction of water droplets during self-cleaning experiments. As shown, the features are accurately produced for each sample with geometric dimensions as indicated in Table 3. Dual-height pillars are evident in images of Sample 3 along with the dual-height pillars/ribs of Sample 4. Each of the sample features appears smooth and with minimal defects from the soft-lithographic process. Information was gathered from SEM images at different magnifications to measure features of interest. Embossed sample 3 is shown in Fig. 14b, with the top views tilted at 45° and not tilted. Upon careful comparison, the features have similar characteristics as with the PDMS molded Sample 3. There appears to be fine strands of presumably PMMA material on select pillars, which are believed to be remnants of the hot embossing procedure created during the de-molding step.

6.2. Anti-biofouling

To understand the antifouling effectiveness of the various samples, a series of bioassay experiments were conducted. Samples were subjected to bioassay experimentation to determine their antifouling efficiency. Fig. 15 shows light microscope images the *E. coli* colonized on a flat sample at two magnifications. As shown, the *E. coli* are long cylindrically shaped microorganisms and tend to cluster in groups and align themselves from head to tail.

Presented for flat PDMS samples at the two concentrations and two incubation times are light microscope images shown in Fig. 16a (with *E. coli* highlighted by the SPIP software). Shown in Fig. 16b are the tabulated results of percentage coverage area of the *E. coli* on the surfaces for 1/100 and 1/10 concentrations at 2 and 4 h incubation time. The greatest coverage occurs at the higher incubation time as *E. coli* growth occurs exponentially over time. Furthermore, as expected, the lower concentration does not

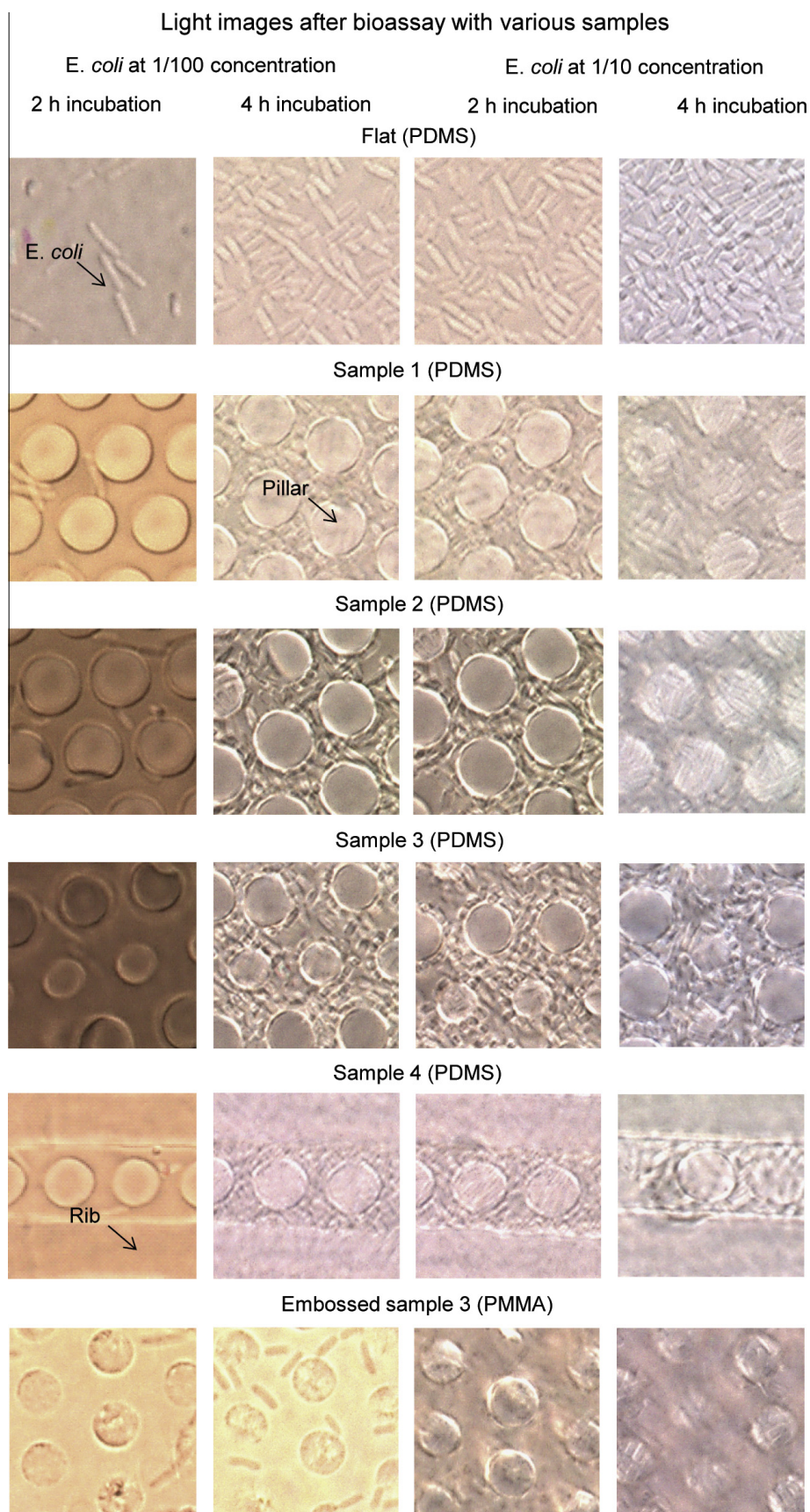


Fig. 17. Light microscope images using oil immersion techniques at 1000 $\times$ magnification of samples after bioassay experiments using *E. coli*. Presented are images of flat PDMS, Samples 1–4, and Embossed sample 3 with two *E. coli* concentrations and two incubation times. Long and narrow cylindrically shaped objects are the *E. coli*. Antifouling analysis was conducted using the images and particle analysis software to quantify effectiveness.

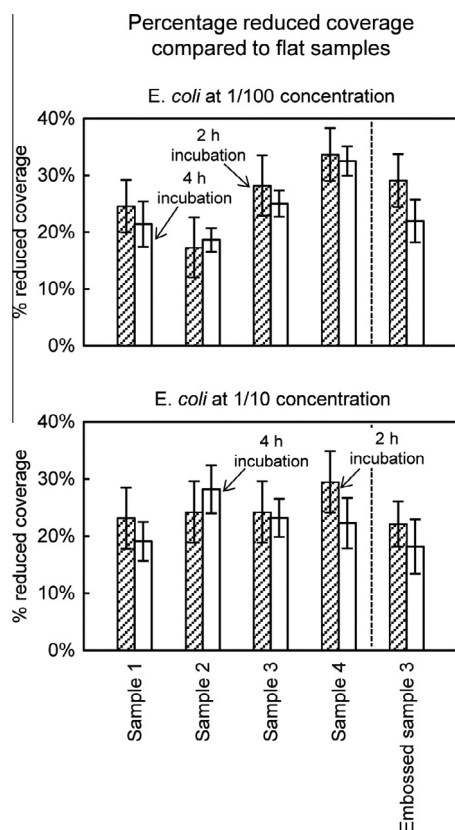


Fig. 18. Anti-biofouling results from *E. coli* bioassay experiments. Results are presented for flat PDMS, Samples 1–4, and Embossed sample 3. The percentage reduced coverage is calculated from the flat PDMS control samples. Sample 4 appears to provide the greatest antifouling efficiency compared to the other samples, which are close in value. It is believed that antifouling efficiency improves with high CA and low CAH. Error bars show ± 1 standard deviation.

provide as much coverage area, however the differences are within 10% for both incubation times. Fig. 17 shows light microscopic images of the flat PDMS control sample, Samples 1–4, and the Embossed sample 3. Tabulated data of percentage reduced coverage compared to the flat control samples are shown in Fig. 18.

As anticipated due to the discontinuities of the various surfaces, Samples 1–4 and the Embossed sample 3 provided antifouling benefit. Antifouling is determined by counting the total number of *E. coli* on the flat control samples and comparing that to the other samples, with results reported in percentage *E. coli* reduced coverage. The greatest antifouling benefit is demonstrated with Sample 4 at 1/100 and 1/10 concentrations at 33% and 29% reduced coverage, respectively. Interestingly, all samples are shown to provide antifouling benefit and are within about 10% of each other's value. In general, antifouling benefit is greater with the 2 h incubation duration as compared to the 4 h duration. It is believed that the longer duration leads to more significant coverage, and the features provide less overall benefit.

Images indicate that *E. coli* are less likely to settle on and colonize the pillar or rib tops, and most often settled in between the features. Interestingly, at 1/100 concentration for 2 h incubation time the Embossed sample 3 provides nearly the same antifouling benefit as Sample 3, with 28% and 29% coverage area reduction, respectively. It should be noted that Sample 3 and Embossed sample 3 performed similarly, indicating minimal differences between samples fabricated from PDMS or PMMA with the fluorchemical release coating. It is possible that the Embossed sample 3 without the fluorchemical coating would not provide as much antifouling benefit. This observation generally holds true for the other concentrations and incubation times. Furthermore, it is believed that high

CA and low CAH will amplify the antifouling abilities of the samples, as it is believed that microorganisms will have more difficulty adhering to the surface during fluid flow.

6.3. Anti-inorganic fouling

To understand the anti-inorganic fouling effectiveness of the various samples, a series of self-cleaning wash experiments are reported from a previous study by Bixler and Bhushan [48]. Presented for the high velocity wash experimentation is the before and after images shown in Fig. 19 for flat PDMS and PMMA control samples, Samples 1–4, and the Embossed sample 3. These images were used with SPIP imaging software to quantify the percentage of particles removed. Tabulated data are shown in Fig. 20 for both the low and high velocity droplet wash experiments.

As expected, the superhydrophobic rice leaf replica outperformed the other samples, and more particles were removed at higher versus lower velocities. Self-cleaning is demonstrated with rice leaf replica and Embossed sample 3 at 95% and 86% contaminant removal, respectively; as compared to uncoated samples at 85% and 70%, respectively [48]. Interestingly, the Embossed sample 3 provides nearly the same self-cleaning ability as the rice leaf replica, even though Embossed sample 3 is not superhydrophobic. Nevertheless, high CA and low CAH is believed to amplify the self-cleaning abilities of the samples, as it is believed that the droplets are able to roll and collect the particles after impact. Furthermore, the nanostructure-coated samples are believed to exhibit lower adhesion forces, suggesting that the particles are easier to remove versus uncoated.

6.4. Wettability

Contact angle measurements were collected by Bixler and Bhushan [48] in both the stream-wise and transverse flow directions, with the maximum values reported. Contact angles in transverse and stream-wise flow are quite different: rice leaf samples show a lower water contact angle when viewed in the stream-wise compared to the cross-stream direction, since the droplets are pinned between the longitudinal grooves.

Actual rice leaf samples with higher contact angles are believed to exhibit Cassie-Baxter wetting where air pockets are trapped beneath the droplet to create superhydrophobicity [63]. Conversely, Sample 1 shows a lower contact angle, presumably due to the Wenzel wetting when water penetrates between the individual asperities. Also as expected, the superhydrophobic rice leaf replica sample exhibits a higher CA than the uncoated samples, showing the effectiveness of the superhydrophobic coating [63]. Furthermore, PDMS Samples 1–4 exhibit hydrophobic contact angles with Sample 3 the highest at 146°. The PMMA samples exhibit lower CA, and for comparison Embossed sample 3 yields a CA of 101°. Since the surface energy of PDMS is lower than PMMA, this trend is expected. Tabulated CA and CAH are shown in Fig. 21. As indicated, actual rice leaf provides the highest CA at 164° with the lowest CAH at 3°, indicating that water droplets are easily repelled [48]. For comparison, maximum water contact angle of actual lotus leaves is 164° [120].

7. Anti-biofouling and anti-inorganic fouling models

Using the experimental and modeling information, designing a new bioinspired antifouling surface is possible by combining shark skin and lotus leaf effects. The rice leaf surface is desirable due to its self-cleaning properties, as well as relatively simple two-dimensional cylindrical pillar geometry. The rice and butterfly wing effect

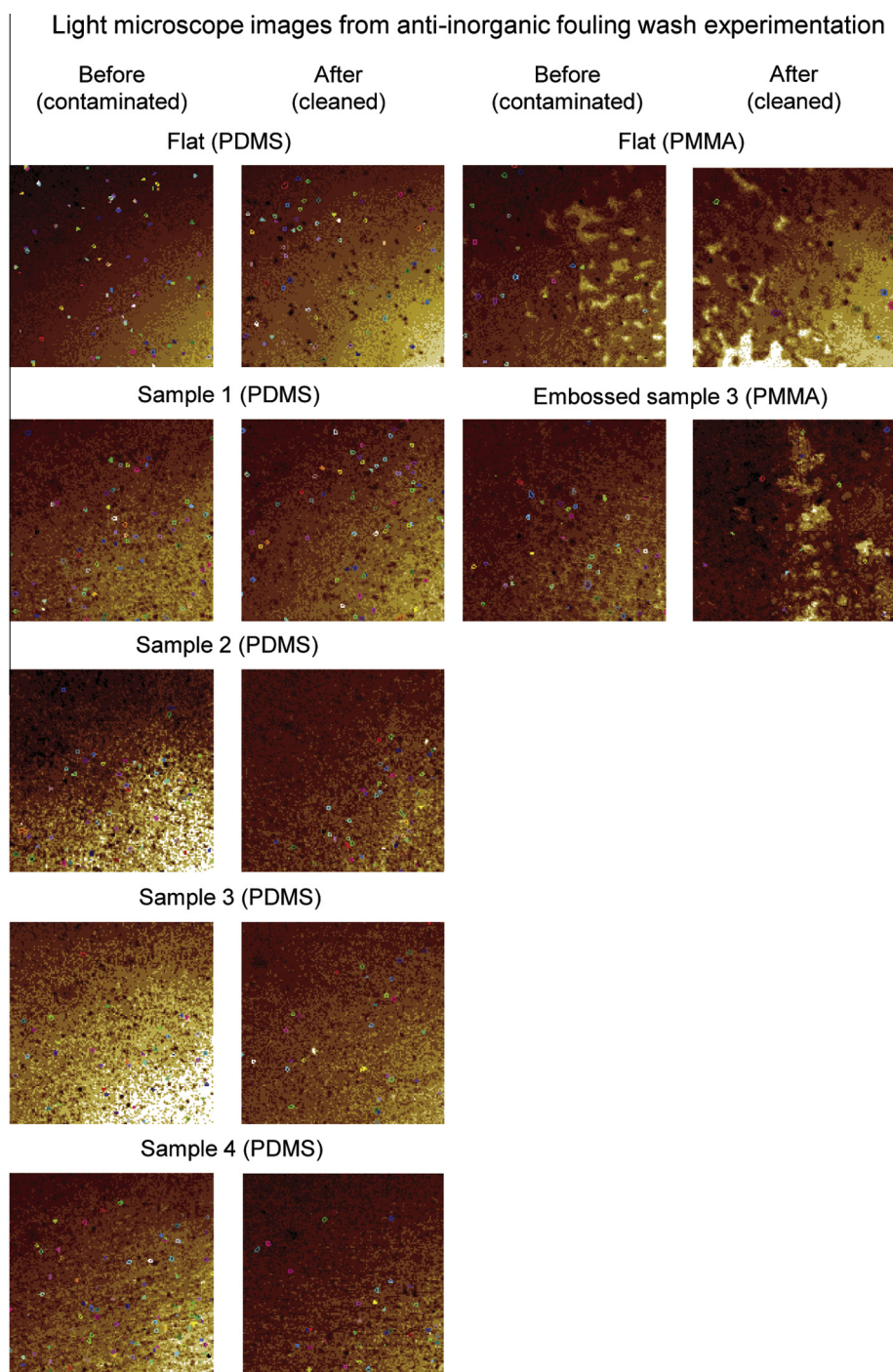


Fig. 19. Light microscope images of samples before and after anti-inorganic fouling wash experimentation, with specs representing silicon carbide (SiC) particles [48]. Results are presented for flat PDMS and PMMA, Samples 1–4, and Embossed sample 3. Self-cleaning analysis was conducted using the images and particle analysis software to quantify particle removal.

may be designed using a uniform micropattern of superhydrophobic low adhesion cylindrical pillars arranged in longitudinal rows.

Various experimental results indicate that rice and butterfly wing effect surfaces provide antifouling benefit, and it is important to understand the mechanisms at work. Shown in Fig. 22 are four cases where *E. coli* shaped and sized microorganisms are exposed to flat and Sample 1 morphologies. As indicated, microorganisms are unable to effectively colonize as well the Sample 1 morphology as compared to the flat control sample. Such surfaces may contain appropriately sized and space discontinuities in the form of pillars. It is believed that microorganisms slightly larger than the gap

spacing between features are less likely to colonize the surface. Furthermore, we believe that the sharp edges of the pillars discourage the settling and eventual colonization of microorganisms on the surface. *E. coli* are shaped and sized such that the edges provide an inconvenient landing zone for settlement, and thus continue the search for an ideal location to colonize. It is believed during this process that many *E. coli* simply expire or do not find a suitable location to permanently attach.

In many cases, the *E. coli* prefer to settle in clusters and around the bases of the pillars, which provide multiple attachment points of contact. When the *E. coli* concentrations are low and the

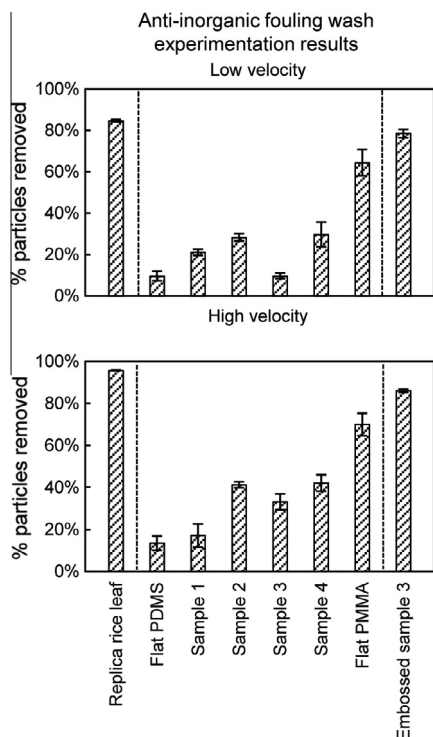


Fig. 20. Anti-inorganic fouling wash experimentation results [48]. Results are presented for rice leaf replica (with superhydrophobic nanostructured coating), flat PDMS and PMMA, Samples 1–4, and Embossed sample 3. The rice leaf replica shows the greatest self-cleaning efficiency followed by the Embossed sample 3. Self-cleaning efficiency improves with high CA and low CAH. Error bars show ± 1 standard deviation.

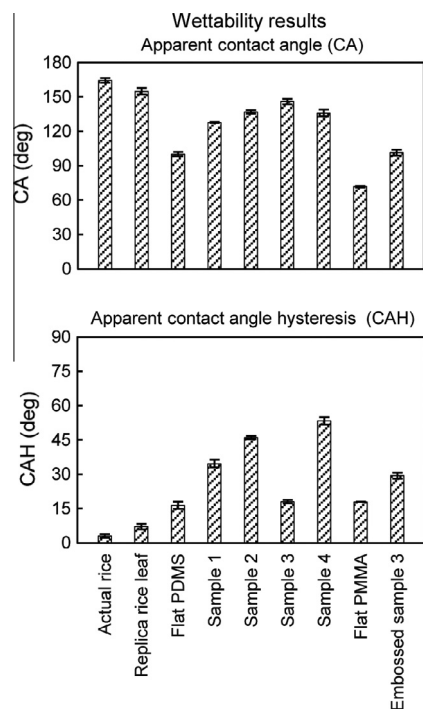


Fig. 21. Results from apparent water contact angle (CA) and contact angle hysteresis (CAH) experiments [48]. Results are presented for actual rice leaf, rice leaf replica (with superhydrophobic nanostructured coating), flat PDMS and PMMA, Samples 1–4, and Embossed sample 3. Self-cleaning efficiency improves with high CA and low CAH. Error bars show ± 1 standard deviation.

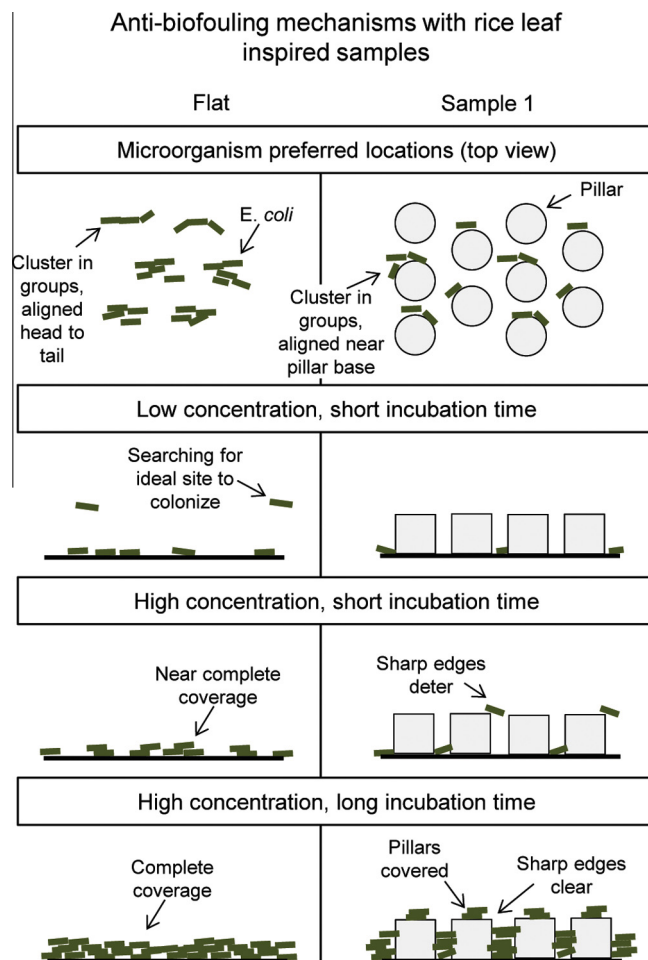


Fig. 22. Anti-biofouling mechanisms with rice leaf inspired samples. Shown are conceptual models based on anti-biofouling experiments using *E. coli* with low to high concentrations and short and long incubation times. Microorganisms prefer to cluster in groups and around the base of pillars. Such microstructured discontinuities and sharp edges are believed to discourage *E. coli* colonization – where they may expire in the process of searching for an ideal landing site. Similar anti-biofouling mechanisms are expected with microorganisms of various shapes and sizes.

incubation times are short, only a few *E. coli* are able to colonize the sample surface. However, when the concentrations are high and the incubation times are long, the *E. coli* eventually will cover over the pillars and create a biofilm. Nevertheless, the discontinuities found with the sharp edged micro-sized pillars are demonstrated to provide antifouling benefit and extend the time at which a particular surface becomes colonized. Such surfaces may be useful in applications where fluid flow is present (i.e. continue to wash away fouling) as well as where cleaning is routine (i.e. in hospitals). The height of the pillars also seems to play a role in anti-fouling, as shorter pillars found with Samples 1 and 3 were covered with *E. coli* sooner than the taller features with Samples 2 and 4. Nevertheless, such mechanisms are believed to be effective against microorganisms of various shapes and sizes. It is believed that postponing microorganism colonization time could aid in the preventing and spread of infectious diseases. This is true since often surfaces are cleaned periodically and therefore will be less likely to harbor a hazardous biofilm.

Contact angle is an important attribute for antifouling of both biofouling and inorganic fouling. As shown in Fig. 23, such foulers may be removed from a surface by water droplets or fluid flow when the rice and butterfly wing effect is present. Close examination of the liquid–solid–air interface reveals that the Wenzel

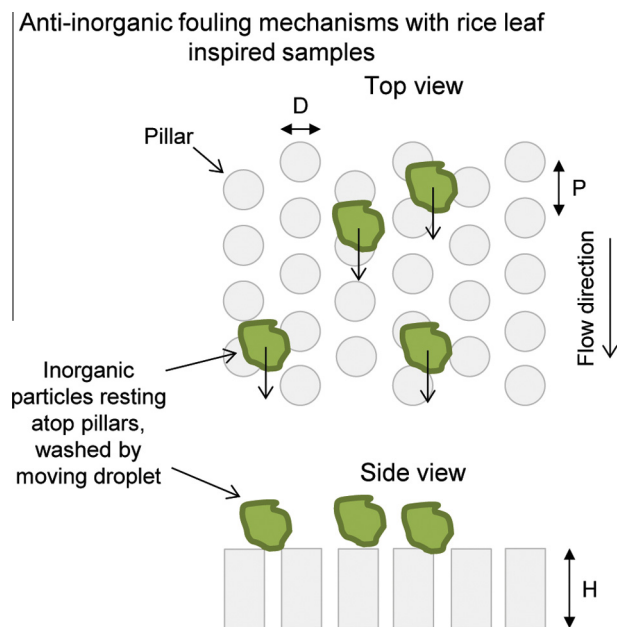


Fig. 23. Anti-inorganic fouling mechanisms with rice leaf inspired samples (adapted from Bixler and Bhushan [48,63]). Model shown utilizes uniformly spaced micro-sized single height pillars in a hexagonal array. Inorganic fouling may be washed away in fluid flow via self-cleaning when coupled with high CA and low CAH.

regime does not contain an air pocket unlike the Cassie-Baxter regime, although both may lead to self-cleaning due to the lotus and pitcher plant effects, respectively. This difference, due to surface roughness and chemistry, influences the surface wettability since the air pocket encourages a larger contact angle and smaller CAH. In general, it is desirable to mimic the lotus effect for self-cleaning, and when combined with the shark skin effect leads to the improved rice and butterfly wing effect. Such a pattern is believed to encourage self-cleaning and directional fluid control, which is believed to provide antifouling benefit.

8. Conclusions

Novel bioinspired microstructured surfaces are shown to provide promising antifouling benefit against biological and inorganic foulers. The samples combine the desirable shark skin (antifouling) and lotus leaf (self-cleaning) effects – which is also called the rice and butterfly wing effect. We examined samples fabricated by photolithography, soft-lithography, and hot-embossing procedures with inspiration from the relatively simple micropapillae morphology of rice leaves.

This study focused on anti-biofouling of *E. coli* microorganisms using various concentrations and incubation times. We also report anti-inorganic fouling results from wash experiments using SiC simulated dirt particles. Measurements and images from light microscopy and SEM are included to provide qualitative and quantitative information on surface features and antifouling effectiveness. The greatest anti-biofouling benefit is shown with the Sample 4 at 33% coverage area reduction, although other samples are within 10%. Additionally, Embossed sample 3 shows similar antifouling performance as Sample 3, at 28% and 29% coverage area reduction, respectively. These findings are promising for scaled-up production of new low-cost embossed films that may be applied to a variety of curved and flat surfaces. The greatest self-cleaning is shown with the lotus effect samples; where the maximum

contaminant removal is reported with Embossed sample 3 and rice leaf replica at 86% and 95%, respectively.

We believe that the sharp edges of the pillars discourage the settling and eventual colonization of microorganisms on the surface. *E. coli* are shaped and sized such that the edges provide an inconvenient landing place for such settlement, and thus continue the search for an ideal location to colonize. It is believed during this process that many *E. coli* that do not attach remain metabolically active throughout the experiment. As such, the gentle rinse from bioassay experiments removes any unattached floaters leaving only the *E. coli* that did attach during the bioassay. In many cases, the *E. coli* prefer to settle in clusters and around the bases of the pillars. Furthermore, self-cleaning results indicate that samples with high CA and low CAH provide the greatest anti-inorganic fouling benefit, as evidenced in nature with actual rice leaves and butterfly wings.

Additional research may provide greater insight into antifouling optimization, including more analysis with other microorganisms and contaminant particles of varying shapes and sizes. In all, this investigation has successfully developed and characterized new bioinspired antifouling surfaces; serving as inspiration in medical, marine, and industrial applications.

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